

Rahul Kothagundla

+91-8500157826 | rahulkothagundla2002@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

Professional Summary

Infrastructure Software Engineer with 1.5+ years building distributed systems on Azure at enterprise scale. Delivered 70% performance improvements on production workloads processing 50K+ daily operations, designed self-healing automation frameworks and maintained operational excellence. Expertise in distributed systems architecture, cloud infrastructure automation and production reliability. Seeking Software Engineer role at Microsoft Azure to build foundational compute infrastructure serving the global cloud platform.

Technical Skills

Languages:	C++, C, Python, Java, SQL
Frameworks & Libraries:	Apache Kafka, Apache Spark, FastAPI, PySpark, Docker, REST APIs
Databases & Storage:	PostgreSQL, Redis, SQL Server, Azure SQL Database
Cloud & Infrastructure:	Azure (Databricks, Compute, Storage), AWS, Docker Compose, CI/CD, Git, Linux
Core Competencies:	Distributed Systems, System Design, Data Structures & Algorithms

Professional Experience

Tata Consultancy Services <i>Systems Engineer</i>	Jun 2024 – Present <i>Hyderabad</i>
- Engineered distributed data processing infrastructure on Microsoft Azure Databricks handling 50K+ daily operations, re-architecting computational pipeline to offload heavy processing and reduce system reload time by 70% (15min → 7min). - Designed production grade self monitoring automation framework with automated failure detection, error handling and recovery mechanisms, maintaining data integrity across distributed pipeline stages serving enterprise users. - Built end-to-end ETL automation pipelines from scratch implementing multi-stage validation, transformation logic and fault-tolerant design, enabling real-time analytics. - Established infrastructure documentation standards covering system architecture, troubleshooting procedures and operational runbooks, reducing engineer onboarding by 90% (7 weeks → 4 days). Earned 5-star rating.	
Atomstate Technologies <i>Machine Learning Engineer</i>	Jun 2023 – Sep 2023 <i>Hyderabad</i>
- Built and deployed NLP based sentiment analysis service with automated data pipeline architecture, processing unstructured text at scale using distributed inference framework. - Developed predictive modeling infrastructure processing large datasets for analytics platform, implementing automated data ingestion and model serving layers.	
Zummit Infolabs <i>Junior Data Scientist</i>	Nov 2022 – Mar 2023 <i>Remote</i>
Applied NLP techniques to build analytical models transforming unstructured data into structured business intelligence with automated extraction pipelines.	

Technical Projects

Real-Time Market Data Streaming Platform Python, Kafka, Spark, PostgreSQL, Redis, Docker
- Engineered distributed streaming infrastructure processing live stock market data with sub-30s latency, handling 1000+ events/second using Apache Kafka with 3-partition topic configuration and fault tolerant producer/consumer design. - Implemented Spark Structured Streaming with exactly once processing semantics, PostgreSQL with optimized composite indexes and Redis caching layer reducing database load by 60% while maintaining data consistency. - Achieved 99.9% system uptime through containerized microservices architecture with automated health checks, restart policies and comprehensive monitoring infrastructure for tracking pipeline health and performance metrics. - Designed for horizontal scalability with stateless services, distributed state management and observability instrumentation ready for production deployment at cloud scale.

Education

B.Tech in Computer Science And Engineering – Data Science VNR VJIET	2020 – 2024 CGPA: 8.55/10
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Certifications & Problem Solving

AWS Cloud Quest: Generative AI Practitioner (Jan 2026)
Amazon ML Summer School 2023
HackerRank Problem Solving
ISRO Geo Data Processing with Python